Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	24 October 2026
Team ID	
Project Name	Incident Lifecycle Automation in ServiceNow
Maximum Marks	4 Marks

Technical Architecture

Every incident moves through four stages on one ServiceNow cloud instance: create and classify, knowledge and reassignment, change and child incident, and resolution with knowledge creation.

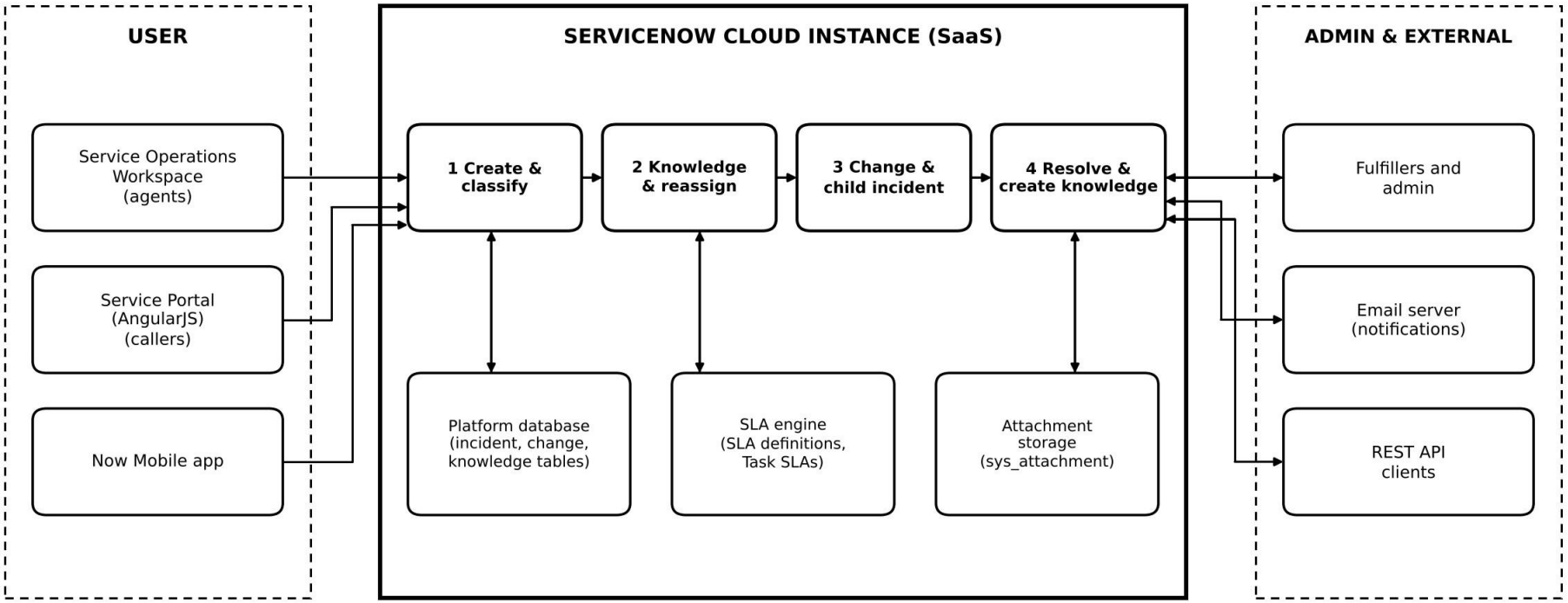


Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Service Operations Workspace for agents and technicians to create, classify and work incidents. Web portal and mobile app for callers.	Service Operations Workspace, Service Portal (HTML, CSS, JavaScript, AngularJS), Now Mobile app
2	Application Logic-1	Incident creation and classification: form fields, category and subcategory, urgency, service, service offering and configuration item.	Incident module, UI Policies, Client Scripts, Business Rules
3	Application Logic-2	Knowledge integration: Agent Assist suggests articles, which are marked helpful and attached. Knowledge article creation from the resolved incident.	Agent Assist, Knowledge Management (IT knowledge base, Standard template)
4	Application Logic-3	Reassignment and escalation: assignment groups (Service Desk, Network), Watch list, Work notes list and On Hold state.	Assignment Groups, Roles, Watch list, Notifications
5	Application Logic-4	Change and related incidents: emergency change request from the incident, child incidents and related lists.	Change Management, Parent-Child Incidents, Related Lists
6	Application Logic-5	SLA tracking: Task SLAs visible in Related Records of the incident.	SLA Definitions, Task SLAs
7	Database	Incident, change, knowledge and SLA data in relational tables (the incident table extends task).	ServiceNow platform relational database
8	Cloud Database	Database service hosted on the cloud instance.	ServiceNow cloud instance database (Personal Developer Instance)
9	File Storage	Attachments on incident records.	ServiceNow attachment storage (sys_attachment)

10	Configuration Data	Services, service offerings and configuration items used to classify incidents.	CMDB (cmdb_ci_service, service_offering)
11	External API-1	Email notifications to watchers and callers.	Notifications, Inbound Email Actions, SMTP/IMAP
12	External API-2	Optional integration of third-party systems with incident data.	REST API / REST Message (JSON)
S.No	**Component**	**Description**	**Technology**
13	Machine Learning Model	Not used in this project. Predictive Intelligence and Virtual Agent are possible later extensions.	Not used
14	Infrastructure (Server / Cloud)	Application deployment on the cloud. Local server configuration: not required. Cloud server configuration: ServiceNow-hosted instance.	ServiceNow cloud instance (SaaS)

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Open-source front-end frameworks used in the portal.	AngularJS, Bootstrap, jQuery
2	Security Implementations	Access control, authentication, user impersonation for role testing, encryption and auditing.	ACLs, roles and groups, HTTPS/TLS, SSO (SAML) and MFA, audit and history logs
3	Scalable Architecture	Three-tier cloud architecture: presentation, application and database.	ServiceNow multi-tier cloud architecture
4	Availability	Data-center replication, failover and managed backups.	ServiceNow cloud infrastructure
5	Performance	Indexing, caching, scheduled jobs and asynchronous business rules.	DB indexing, platform caching, scheduled jobs, async business rules